

Semi-Supervised Machine Learning

Complete Algorithm Family Tree

A comprehensive reference covering every major algorithm family, variant, and technique in semi-supervised learning

15	Major Families
130+	Algorithms & Variants
5	Application Domains

Table of Contents

#	Family	Coverage
1.	Generative Model-Based Methods	10 algorithms
2.	Graph-Based Methods	16 algorithms
3.	Self-Training Methods	11 algorithms
4.	Co-Training Methods	10 algorithms
5.	Consistency Regularization Methods	13 algorithms
6.	Entropy Minimization Methods	6 algorithms
7.	Semi-Supervised SVMs (S3VM/TSVM)	8 algorithms
8.	Semi-Supervised Clustering Methods	10 algorithms
9.	Active Learning & Query Strategies	10 algorithms
10.	Contrastive & Self-Supervised Pretraining	12 algorithms
11.	Semi-Supervised Regression Methods	6 algorithms
12.	Sequence & Structured Prediction	9 algorithms
13.	Deep Semi-Supervised Learning (Modern)	12 algorithms
14.	Semi-Supervised Dimensionality Reduction	6 algorithms
15.	Federated & Privacy-Preserving SSL	3 algorithms

Introduction to Semi-Supervised Learning

Semi-supervised learning (SSL) occupies the space between fully supervised learning — where every training example carries a label — and fully unsupervised learning. In most real-world settings, collecting large amounts of raw data is inexpensive, but obtaining reliable annotations requires significant human effort and domain expertise. SSL methods address this mismatch by leveraging the structure present in unlabeled data to augment the signal from a small labeled set.

The core assumptions underpinning semi-supervised learning are: (1) the **smoothness assumption** — nearby points in input space likely share the same label; (2) the **cluster assumption** — points within the same cluster tend to share a label; and (3) the **manifold assumption** — data lies on a low-dimensional manifold embedded in high-dimensional space, and the decision boundary should follow that manifold.

This document catalogues **every major algorithm family and variant** in semi-supervised machine learning, organised into 15 families. Each entry includes the algorithm's core mechanism, key design decisions, and notable variants, providing a comprehensive reference for practitioners and researchers.

Core SSL Assumptions

Assumption	Description	Methods That Use It
Smoothness	Close inputs → same label	Label Propagation, Graph Methods
Cluster	Same cluster → same label	S3VM, Entropy Minimization
Manifold	Data lies on low-dim manifold	Manifold Regularization, GNN
Low Density Separation	Decision boundary in low-density region	TSVM, S3VM, VAT

1. Generative Model-Based Methods

These methods model the joint distribution $P(X,Y)$ or the data likelihood $P(X)$. Semi-supervised learning is achieved by using unlabeled data to better estimate the data-generating distribution, then combining that with labeled supervision.

■ Gaussian Mixture Models (GMM) — Semi-Supervised

Models data as a mixture of Gaussians. Labeled data pins class–component assignments; unlabeled data refines the mixture parameters via EM. The resulting posterior is used for classification.

■ Expectation-Maximization (EM) for Semi-Supervised Learning

A general framework: the E-step infers soft labels for unlabeled points; the M-step re-estimates model parameters using both labeled and pseudo-labeled data. Converges to a local likelihood maximum.

■ Naive Bayes — Semi-Supervised (EM variant)

Applies EM to the Naive Bayes generative model. Unlabeled documents update the word-probability tables in each EM iteration, improving the class conditional estimates with minimal labeling cost.

■ Variational Autoencoders (VAE) — Semi-Supervised (Kingma et al., 2014)

Latent-variable generative model. The M2 model introduces a discrete label variable alongside a continuous latent code. The ELBO is augmented with a supervised classification term for labeled examples.

■ Generative Adversarial Networks (GAN) — Semi-Supervised (SGAN)

The discriminator is trained to output $K+1$ classes (K real classes + 1 fake). Fake samples from the generator act as an implicit regulariser, sharpening the real-class decision boundaries.

■ Semi-Supervised GAN (SSGAN / Feature Matching GAN)

Adds a feature-matching loss between real and generator-produced features, stabilising training and improving representation quality for downstream classification on labeled subsets.

■ Latent Dirichlet Allocation (LDA) — Semi-Supervised

Topic model extended so that document labels partially constrain topic distributions. Labeled documents guide topic inference; unlabeled ones benefit from the richer topic structure discovered jointly.

■ Hidden Markov Models (HMM) — Semi-Supervised

EM is applied to the HMM with unlabeled sequences contributing to the transition and emission parameter updates, widely used in speech recognition and biological sequence labeling.

■ Deep Generative Models — Auxiliary Deep Generative Model (ADGM)

Introduces an auxiliary latent variable to create a richer approximate posterior, improving the tightness of the variational lower bound in the semi-supervised VAE setting.

■ Ladder Variational Autoencoder (LVAE) — Semi-Supervised

Hierarchical VAE with top-down inference corrections. The hierarchical latent structure captures multi-scale data features, and the semi-supervised extension ties the top latent to class labels.

2. Graph-Based Methods

Graph-based methods construct a similarity graph over all data points (labeled + unlabeled). Labels propagate or regularise smoothly along high-density graph edges, exploiting the manifold assumption.

■ Label Propagation (LP)

Iteratively spreads labels from labeled nodes to neighbors weighted by similarity. Each node's label distribution is a weighted average of its neighbors; labeled nodes clamp their true labels throughout.

■ Label Spreading (LS)

A normalized variant of LP that allows labeled nodes to adjust their own labels slightly, controlled by a clamping parameter α . More robust to label noise than standard LP.

■ Gaussian Fields and Harmonic Functions (Zhu et al., 2003)

Assigns labels via harmonic functions on the graph: the label at each unlabeled node minimises a quadratic energy (Laplacian regularisation) with labeled nodes as boundary conditions.

■ Manifold Regularization (Belkin et al., 2006)

Augments any supervised loss with a graph Laplacian penalty that penalises label changes along the data manifold. Implemented efficiently with kernel methods (Laplacian SVM, Laplacian RLS).

■ Tikhonov Regularization on Graphs

Uses the graph Laplacian as a regulariser in a Tikhonov-style objective. Provides a principled Bayesian interpretation as a Gaussian process prior encoding smoothness on the graph.

■ Graph Neural Networks (GNN) — Semi-Supervised (Kipf & Welling, 2017)

Graph Convolutional Networks (GCN) perform spectral convolution on the adjacency matrix, learning node embeddings that aggregate neighborhood information. Only a small fraction of nodes need labels.

■ Graph Attention Networks (GAT) — Semi-Supervised

Replaces fixed graph convolution weights with attention coefficients learned per edge, allowing the model to focus on the most informative neighbors for each node during message passing.

■ GraphSAGE — Semi-Supervised

Inductive GNN that learns aggregator functions (mean, LSTM, pooling) over sampled neighborhoods, enabling generalization to unseen nodes without retraining the full graph.

■ Graph Convolutional Matrix Completion

Uses GCN encoder on the bipartite user-item graph for matrix completion, leveraging both graph structure and side-information in a semi-supervised manner for recommendation systems.

■ Deep Graph Infomax (DGI)

Maximises mutual information between local patch representations and a global graph summary. The self-supervised DGI objective provides rich node features that greatly reduce labeled data requirements.

■ Contrastive Multi-View Graph Clustering (CMGC)

Combines contrastive learning across multiple graph views with graph clustering objectives, jointly leveraging structure and feature spaces in a semi-supervised or unsupervised manner.

■ S4GNN — Scalable Semi-Supervised GNN

Decouples feature propagation from neural network training to scale to large graphs, using pre-computed aggregated features and a lightweight MLP classifier.

■ Correct & Smooth (C&S;)

Post-processing pipeline: first train a base model on labeled nodes, then use label propagation to correct errors and smooth predictions across the graph structure.

■ APPNP (Approximate Personalized Propagation of Neural Predictions)

Separates the neural feature transformation from the propagation step using personalized PageRank, allowing deeper propagation without over-smoothing.

■ GRAND — Graph Random Neural Networks

Applies random propagation (DropNode augmentation) during training and enforces consistency across multiple propagation instances via a consistency regularisation loss.

■ Geom-GCN — Geometric Aggregation for Graph Neural Networks

Maps nodes to a latent geometric space and aggregates in structural neighborhoods defined there, addressing the limitation of GCNs that ignore node structural roles.

3. Self-Training Methods

Self-training (also called self-learning or decision-directed learning) trains a model on labeled data, assigns pseudo-labels to high-confidence unlabeled samples, and iterates. It is a wrapper method compatible with any base classifier.

■ Vanilla Self-Training (Scudder, 1965)

Train on labeled set → predict on unlabeled set → add high-confidence predictions as pseudo-labels → retrain. Repeat until convergence. The simplest and most widely deployed semi-supervised approach.

■ Pseudo-Label (Lee, 2013)

In neural network training, uses the class with maximum softmax probability as the pseudo-label for unlabeled examples during each mini-batch update. Gradually increases unlabeled data contribution via a ramp-up schedule.

■ Noisy Student Training (Xie et al., 2020)

Teacher generates pseudo-labels on unlabeled data; student is trained with strong augmentations (noise) applied to both labeled and unlabeled inputs. Iterative process scales to billions of images.

■ Mean Teacher (Tarvainen & Valpola, 2017)

The teacher is an exponential moving average (EMA) of the student's weights. Consistency loss between student and teacher predictions on augmented unlabeled inputs reduces confirmation bias compared to vanilla self-training.

■ Temporal Ensembling (Laine & Aila, 2017)

Maintains an EMA of past network predictions for each unlabeled sample. The consistency loss encourages current predictions to match the accumulated ensemble, acting as a form of temporal model averaging.

■ Self-Training with Noisy Labels (STNL)

Combines pseudo-labeling with noise-robust loss functions (e.g., symmetric cross-entropy, GCE loss) to mitigate the effect of incorrect pseudo-labels accumulated during self-training iterations.

■ Tri-Training

Trains three classifiers on bootstrap samples of the labeled set. An unlabeled sample is added to a classifier's training set if the other two classifiers agree on the label, providing a built-in disagreement filter.

■ Democratic Co-Learning

Extension of tri-training to more than three learners with different representation types. Labels are accepted by voting majority, and class-balancing constraints prevent degenerate solutions.

■ FEAST — Feature-Ensemble-based Self-Training

Ensembles predictions from multiple feature views of the same data, selecting pseudo-labels where all views agree to minimise noise injection into the self-training loop.

■ AdaBoost-based Self-Training

Uses AdaBoost as the base learner in self-training, re-weighting examples at each iteration so that hard-to-classify unlabeled samples receive more attention in subsequent rounds.

■ S4L — Self-Supervised Semi-Supervised Learning

Augments supervised training with self-supervised auxiliary tasks (rotation prediction, exemplar learning) to better utilise unlabeled data through shared representation learning.

4. Co-Training Methods

Co-training exploits multiple conditionally independent views of the data. Each learner trains on one view and labels unlabeled examples for the other learner. The diversity across views drives the semi-supervised improvement.

■ Blum & Mitchell Co-Training (1998)

The original co-training algorithm. Two classifiers are trained on two independent feature views (e.g., text content and hyperlinks for web pages). Each classifier adds its most confident unlabeled predictions to the other's labeled training pool.

■ Co-EM

Combines the co-training framework with EM by using the posterior from one view as a soft label for the other, iterating until convergence. Smoother than hard co-training label assignments.

■ Co-Training with Disagreement (Dasgupta et al.)

Formalises the theory behind co-training: classifiers should agree on labeled data and minimise disagreement on unlabeled data, subject to each classifier having low error on its own view.

■ Multiview Co-Training

Generalises co-training to more than two views. Each view trains a separate model; pseudo-labels are generated by consensus or vote across all view models, reducing variance in the pseudo-label quality.

■ Co-Training with SVMs

Employs SVMs as the base classifiers in each view, leveraging margin maximisation to select the most informative unlabeled samples to exchange between views.

■ Co-Training for Deep Networks (DCTN)

Applies co-training to deep neural networks using different network architectures or different augmentation strategies as the two 'views', circumventing the need for naturally disjoint feature sets.

■ Co-Training via Contrastive Learning

Replaces one view with contrastive self-supervised representations, enabling co-training even when only a single feature set is available.

■ Asymmetric Co-Training

Relaxes the symmetry requirement: one stronger model labels for a weaker model (teacher–student flavour), useful when views differ significantly in quality.

■ Co-Training with Neural Networks (CoTraining-NN)

Uses two neural networks that share a common backbone but have different prediction heads or dropout patterns as the two co-training classifiers.

■ Co-Forest

Random forest variant of co-training: an ensemble of trees constitutes each view, and pseudo-labels with high confidence across trees are exchanged between the two forest ensembles.

5. Consistency Regularization Methods

These methods enforce that model predictions should be invariant to small perturbations of the input (data augmentations, noise, dropout). Only the consistency loss on unlabeled data is required, making them highly practical.

■ Virtual Adversarial Training (VAT) — Miyato et al., 2018

Computes the adversarial perturbation that maximally changes the model's prediction (without using the true label), then minimises the KL divergence between clean and perturbed predictions for both labeled and unlabeled data.

■ Pi-Model (Laine & Aila, 2017)

Applies two stochastic augmentations to the same unlabeled input and minimises the mean-squared error between the two resulting predictions. Augmentations include Gaussian noise, translation, and random flipping.

■ Interpolation Consistency Training (ICT)

Uses Mixup interpolations of unlabeled samples as augmentations. The consistency target is the Mixup of the model's predictions on the two original samples, combining data augmentation and consistency.

■ UDA — Unsupervised Data Augmentation (Xie et al., 2020)

Uses RandAugment (vision) or back-translation (NLP) as strong augmentations for the consistency loss. The sharper augmentation creates a harder regularisation target, substantially improving accuracy.

■ FixMatch (Sohn et al., 2020)

Combines pseudo-labeling with consistency: a weak augmentation produces a pseudo-label (thresholded by confidence); a strong augmentation produces the prediction to be trained against the pseudo-label. Highly effective benchmark method.

■ FlexMatch

Extends FixMatch with curriculum pseudo-labeling: the confidence threshold is dynamically adjusted per class based on the model's current learning status, addressing the imbalanced threshold problem.

■ FreeMatch

Uses a self-adaptive thresholding strategy based on the model's historical average confidence rather than a fixed threshold, further improving on FlexMatch's dynamic approach.

■ SoftMatch

Replaces hard thresholding with a soft Gaussian weighting function over confidence scores, retaining more pseudo-labeled samples while down-weighting low-confidence ones.

■ MixMatch (Berthelot et al., 2019)

Unifies consistency regularisation, label guessing, and MixUp. Produces a sharpened pseudo-label as the average of K augmented predictions; MixUp is applied to the combined labeled + pseudo-labeled set.

■ ReMixMatch (Berthelot et al., 2020)

Extends MixMatch with two innovations: distribution alignment (adjusts pseudo-label marginals to match the labeled set marginals) and augmentation anchoring (uses a weakly augmented sample to generate the pseudo-label target).

■ DivideMix

Models the loss distribution of noisy labeled data as a two-component GMM (clean vs. noisy), then applies MixMatch separately on the clean partition and the noisy partition as unlabeled data.

■ SimMatch

Propagates pseudo-labels using both semantic similarity (feature space) and instance-level similarity, creating a more accurate pseudo-label distribution than single-view methods.

■ SemiFormer — Semi-Supervised Vision Transformer

Applies consistency regularisation to Vision Transformers (ViT), adapting strong augmentation pipelines (AugReg, MoCo-style augmentations) to the attention-based architecture.

6. Entropy Minimization Methods

Entropy minimisation encourages the model to make confident (low-entropy) predictions on unlabeled data, pushing decision boundaries away from dense data regions toward low-density separators.

■ Entropy Minimization (Grandvalet & Bengio, 2005)

Adds the conditional entropy $H(Y|X)$ on unlabeled data as a regulariser to the supervised cross-entropy loss. Drives the model to commit to a single class for each unlabeled point.

■ Information Maximization (IMSAT)

Maximises mutual information between inputs and predictions while applying VAT-based regularisation. Ensures globally high entropy (diverse predictions) but locally low entropy (confident per-sample predictions).

■ Minimum Entropy Regularization (MER)

Bayesian perspective on entropy minimisation: treats the entropy term as a prior that encourages deterministic classifier boundaries at unlabeled points.

■ Confident Learning (CL)

Estimates the joint distribution between noisy and true labels using confident predictions, enabling simultaneous semi-supervised learning and noise-robust training.

■ SHOT — Source Hypothesis Transfer

Minimises entropy and enforces diversity using information maximisation when adapting to a target domain with no target labels — a form of entropy-minimisation for domain adaptation.

■ Mutual Information Neural Estimation (MINE) for SSL

Uses neural estimators of mutual information as an objective for semi-supervised feature learning, maximising input–output MI on unlabeled data to learn compact discriminative representations.

7. Semi-Supervised Support Vector Machines (S3VM / TSVM)

These methods extend the SVM maximum-margin principle to unlabeled data by seeking a decision hyperplane that has large margin and passes through low-density regions of the input space.

■ Transductive SVM (TSVM) — Vapnik, 1998

Optimises the SVM objective over labeled data while jointly assigning ± 1 labels to unlabeled data to maximise the margin over all points. Solved via a combinatorial search or continuous relaxation.

■ S3VM (Semi-Supervised SVM) — Bennett & Demiriz, 1998

Formulates a mixed-integer program to find the maximum-margin hyperplane treating unlabeled labels as free variables. Various relaxation strategies (DC programming, gradient descent) are used in practice.

■ SVMlin

Efficient linear TSVM that scales to large datasets via an alternating optimisation between labeled SVM training and unlabeled label re-assignment using gradient-based updates on the linear model.

■ LapSVM (Laplacian SVM)

Combines the SVM objective with a Laplacian graph regulariser (manifold regularisation), encoding both the large-margin and geometric smoothness constraints in a single kernel learning problem.

■ DS3VM — Deep Semi-Supervised SVM

Replaces the linear SVM kernel with deep feature representations, jointly training the deep network and the SVM objective using unlabeled samples to encourage low-density separation.

■ Universum TSVM

Augments TSVM with Universum examples (samples that do not belong to any training class) to provide additional prior knowledge about the decision boundary, improving generalisation.

■ Fast TSVM (Collobert et al.)

Uses a concave–convex procedure (CCCP) to efficiently solve the TSVM optimisation by breaking the non-convex problem into a sequence of convex QP subproblems.

■ Robust TSVM

Incorporates label noise tolerance into the TSVM formulation by using robust loss functions (e.g., ramp loss) that down-weight the influence of outliers and mislabeled examples.

8. Semi-Supervised Clustering Methods

These methods incorporate labeled constraints or class information into clustering algorithms, bridging unsupervised clustering and supervised classification.

■ **Constrained k-Means (COP-k-Means)**

Extends k-means with must-link (same cluster) and cannot-link (different cluster) constraints derived from labeled pairs. Enforces constraints during the assignment step to guide cluster formation.

■ **Seeded k-Means / Constrained k-Means (Basu et al., 2002)**

Initialises cluster centroids using labeled examples and enforces cluster membership for labeled points throughout the EM iterations, linking cluster identity to class labels.

■ **Semi-Supervised EM Clustering**

EM algorithm for GMM clustering augmented with labeled component assignments. Labeled data determines which Gaussian component a point belongs to; unlabeled data refines all component parameters.

■ **Semi-Supervised Spectral Clustering**

Modifies the graph Laplacian or eigenvector computation to incorporate pairwise constraints, guiding the spectral embedding toward a structure consistent with known label relationships.

■ **DBSCAN with Constraints**

Augments density-based clustering with must-link and cannot-link constraints, adjusting the neighbourhood density calculation to respect the partial supervision signal.

■ **Semi-Supervised Hierarchical Clustering**

Incorporates labeled pair constraints into agglomerative or divisive hierarchical clustering by modifying the linkage criterion or by post-pruning the dendrogram to satisfy constraints.

■ **MPCKMeans (Metric-Pairwise Constrained K-Means)**

Jointly learns a Mahalanobis distance metric and cluster assignments. Labeled pair constraints guide the metric to bring must-link pairs close and separate cannot-link pairs.

■ **SSSC — Semi-Supervised Subspace Clustering**

Uses label information to guide the sparse subspace clustering optimisation, ensuring that labeled points from the same class reconstruct each other in the self-expressive model.

■ **Deep Adaptive Clustering (DAC)**

Self-supervised image clustering that uses cosine similarity to generate adaptive soft labels, iteratively refining the feature space and cluster assignments.

■ **SCAN — Semantic Clustering by Adopting Nearest Neighbors**

Two-stage: first learns strong features via self-supervised pretraining, then uses nearest-neighbor mining with a small set of labels to semantically separate clusters.

9. Active Learning and Query Strategies in SSL

Active learning selects the most informative unlabeled samples to label, reducing annotation cost. When combined with SSL, it maximises the utility of each labeling query.

■ Uncertainty Sampling

Queries the unlabeled sample for which the current model is least confident (e.g., closest to the decision boundary, highest predictive entropy, or smallest margin between top-2 class probabilities).

■ Query-by-Committee (QBC)

Maintains a committee of diverse models; queries the sample with maximum disagreement among committee members, measuring informativeness by vote entropy or query-by-bagging variants.

■ Core-Set Selection (Sener & Savarese, 2018)

Selects a minimal coreset of unlabeled points such that the maximum distance between any unlabeled point and its nearest selected point is minimised, ensuring geometric coverage.

■ Bayesian Active Learning by Disagreement (BALD)

Selects the sample that maximises the mutual information between model parameters and predictions (information gain), computed approximately via MC-Dropout sampling.

■ Expected Model Change (EMC)

Queries the unlabeled point whose labeling would cause the greatest change to the current model parameters, approximated by the gradient magnitude in the model's parameter space.

■ Expected Error Reduction (EER)

Selects the sample that minimises expected future generalisation error, estimated by integrating over possible labels weighted by the current model's posterior.

■ Diversity-based Active Learning (DAL)

Selects a batch of unlabeled points that are both individually uncertain and mutually diverse (low pairwise similarity), balancing exploration and exploitation.

■ Learning Loss for Active Learning

Attaches a small loss prediction module to the main network that predicts the training loss on each unlabeled sample; the highest predicted-loss samples are queried.

■ VAAL — Variational Adversarial Active Learning

Trains a VAE and adversarial discriminator to distinguish labeled from unlabeled pool distributions, querying points the discriminator classifies as unlabeled (most representative of unexplored regions).

■ Active Learning with MixUp

Combines MixUp-based data augmentation with uncertainty sampling, using the interpolation as a smoothed decision boundary probe to identify the most ambiguous regions.

10. Contrastive and Self-Supervised Pretraining for SSL

Self-supervised methods learn rich representations without labels, which are then fine-tuned with a small labeled set, achieving semi-supervised performance that rivals fully supervised baselines.

■ SimCLR — Simple Framework for Contrastive Learning (Chen et al., 2020)

Augments each image twice and maximises agreement between the two views using NT-Xent loss. The pretrained encoder is fine-tuned on labeled data. Achieves strong semi-supervised performance on ImageNet.

■ MoCo — Momentum Contrast (He et al., 2020)

Maintains a dynamic dictionary of encoded representations via a momentum-updated key encoder and a queue. Enables large-batch contrastive learning without requiring large GPU memory.

■ BYOL — Bootstrap Your Own Latent (Grill et al., 2020)

Contrastive learning without negative pairs. An online network predicts the target network's representation; the target network is updated via EMA of the online network, bootstrapping rich representations.

■ SimSiam (Chen & He, 2021)

Simplifies BYOL by removing the EMA target and using a stop-gradient operation to prevent representation collapse. Demonstrates that self-supervised learning does not require negative pairs or EMA targets.

■ SwAV — Swapping Assignments between multiple Views

Performs online clustering of representations across multiple crops. Swapped prediction: the code assigned from one crop is predicted from the other crop's features without explicit pairwise comparisons.

■ DINO — Self-Distillation with No Labels (Caron et al., 2021)

Teacher–student self-distillation applied to Vision Transformers. The teacher is updated via EMA; features from the last [CLS] token exhibit strong semantic properties useful for semi-supervised tasks.

■ MAE — Masked Autoencoders (He et al., 2022)

Random masking of 75% of image patches; an asymmetric encoder–decoder reconstructs pixel values of masked patches. The pre-trained encoder transfers exceedingly well with few labels.

■ BarlowTwins

Proposes a cross-correlation matrix between two augmented view embeddings to be the identity matrix, decorrelating features and removing redundancy without negative samples or momentum encoders.

■ VICReg — Variance-Invariance-Covariance Regularization

Explicitly controls the variance, invariance, and covariance of learned representations via three complementary loss terms, achieving stable training without collapse.

■ MSN — Masked Siamese Networks (Assran et al., 2022)

Predicts the prototype assignment of a full-view token from a masked view, combining masked image modeling and prototypical contrastive learning for low-shot semi-supervised classification.

■ MoCo v3 / SimCLR v2

Improved versions adding larger projection heads, stronger augmentations, and knowledge distillation for fine-tuning on semi-supervised benchmarks with 1% or 10% labels.

■ PCL — Prototypical Contrastive Learning

Introduces cluster prototypes as anchors in the contrastive loss, bridging instance-level contrastive learning and cluster-level semantic structure for better semi-supervised transfer.

11. Semi-Supervised Regression Methods

Semi-supervised methods for continuous output prediction exploit the distribution of unlabeled inputs to improve regression estimates.

■ Co-Training for Regression

Two regressors are trained on two views; each labels unlabeled examples for the other where prediction confidence (low prediction variance) is high, similar to co-training for classification.

■ Semi-Supervised Ridge Regression (SSRR)

Adds a Laplacian graph regulariser to the ridge regression objective, penalising label variation along the data manifold for smoothness-consistent function estimation.

■ Gaussian Process Regression — Semi-Supervised

Uses unlabeled data to learn a better kernel function (hyperparameter estimation) or to select informative support points, improving the GP posterior with minimal label overhead.

■ Semi-Supervised Random Forest for Regression

Combines supervised regression trees with unsupervised proximity-based label propagation. Unlabeled samples influence the tree structure by participating in split selection criteria.

■ COREG (Co-Regularization for Regression)

Two k-NN regressors on complementary distance metrics label unlabeled examples for each other based on disagreement reduction, creating a consistent co-training paradigm for regression.

■ Semi-Supervised SVR (S3VR)

Extends TSVM/S3VM to regression (ϵ -insensitive loss), seeking a regression function that is flat (low complexity) in labeled regions and passes through low-density regions of the input space.

12. Semi-Supervised Sequence and Structured Prediction

Methods for structured output spaces such as sequences, parse trees, and segmentation maps, where both inputs and outputs have complex dependencies.

■ Semi-Supervised CRF (Conditional Random Field)

Augments the CRF objective with a marginal likelihood term over unlabeled sequences, or uses entropy regularisation on the marginal distribution to encourage confident structured predictions.

■ Semi-Supervised Sequence-to-Sequence (Seq2Seq)

Back-translation: an auxiliary translation model generates pseudo-source sentences from unlabeled target sentences, creating synthetic parallel data for supervised seq2seq training.

■ Cross-View Training (CVT) for NLP

Adds auxiliary prediction heads operating on partial input views (e.g., left context only, right context only). A consistency loss between the full-context head and partial-context heads regularises the encoder.

■ Semi-Supervised Named Entity Recognition (NER)

Combines label propagation on word embeddings, self-training with high-confidence span annotations, and data augmentation (synonym replacement) to handle scarce NER labels.

■ Mean Teacher for Sequence Labeling

Extends the Mean Teacher consistency framework to token-level labeling tasks (POS tagging, NER, chunking) by applying EMA consistency at each token position rather than the whole sequence output.

■ Semi-Supervised Semantic Segmentation

Combines pseudo-label thresholding, CutMix augmentation, and class-balanced sampling to produce pixel-level pseudo-masks for unlabeled images, enabling SSL for dense prediction tasks.

■ CPS — Cross Pseudo Supervision (Chen et al., 2021)

Two segmentation networks with different initialisation impose cross pseudo supervision: each network's confident predictions serve as pseudo-labels for the other network.

■ PC2Seg — Perturb-and-Compare Semantic Segmentation

Uses perturbation-based consistency and a contrastive pixel-to-pixel comparison loss to learn segmentation representations from unlabeled images.

■ Semi-Supervised Object Detection (STAC, Unbiased Teacher)

Applies a teacher–student framework where the EMA teacher generates pseudo-bounding-box annotations for unlabeled images, with data augmentation and filtering strategies to handle noise.

13. Deep Semi-Supervised Learning (Holistic/Recent Methods)

Modern deep learning frameworks that integrate multiple SSL principles (consistency, entropy, pseudo-labels, augmentation) into unified pipelines.

■ EnAET — Self-Trained Ensemble AutoEncoding Transformations

Learns the relationship between random spatial transformations and their encoded parameters to build transformation-equivariant representations useful for SSL fine-tuning.

■ Deep Co-Training (DCT)

Two deep networks with different dropout masks act as two views. Adversarial examples are added to minimise view agreement on unlabeled data, preventing trivial co-training solutions.

■ Dash — Discriminative Few-shot Self-Supervision

Selects pseudo-labeled examples by optimising a data selection criterion that balances prediction confidence and training loss reduction, rather than using a fixed confidence threshold.

■ CSSL — Curriculum Semi-Supervised Learning

Applies a curriculum that starts with easy (high-confidence) unlabeled samples and gradually incorporates harder ones, stabilising training and improving final accuracy.

■ OpenMatch — Open-Set Semi-Supervised Learning

Handles the realistic scenario where unlabeled data contains out-of-distribution (OOD) samples by training an OOD detector jointly with the SSL classifier, filtering OOD samples from pseudo-labeling.

■ SSRE — Semi-Supervised Representation Learning with Energy

Uses energy-based models to assign high energy (low probability) to OOD unlabeled samples and low energy to in-distribution samples, enabling robust SSL in the presence of distribution shift.

■ Class-Imbalanced Semi-Supervised Learning (DARP, ABC)

Addresses the label-distribution mismatch between labeled and unlabeled data. Methods include distribution alignment of pseudo-labels (DARP) and auxiliary balanced classifier heads (ABC).

■ Semi-Supervised Learning with Data Augmentation (RandAugment SSL)

Systematically explores augmentation policies in combination with consistency regularisation, finding that strong augmentation is the single most impactful component for SSL on vision tasks.

■ Semi-Supervised Learning with Pretrained Language Models (BERT, GPT)

Fine-tunes large pretrained transformers (BERT, RoBERTa, T5) on small labeled datasets. The massive unsupervised pretraining effectively subsumes the role of classical SSL objectives for NLP.

■ Semi-Supervised Vision Transformers (SemiViT, DeiT-SSL)

Adapts ViT architectures to semi-supervised settings using patch-level masking (MAE-style) and token consistency regularisation across augmented views.

■ CMatch — Class-Aware Threshold for Semi-Supervised Learning

Dynamically adjusts confidence thresholds per class using estimated class priors from both labeled and unlabeled data, providing a practical solution to class-frequency imbalance in SSL.

■ AdaMatch (Berthelot et al., 2021)

Unifies domain adaptation and semi-supervised learning: applies random logit interpolation and distribution alignment to handle covariate shift between labeled and unlabeled distributions.

14. Semi-Supervised Dimensionality Reduction

These methods use label information to guide dimensionality reduction, producing low-dimensional representations that are both compact and discriminative.

■ Semi-Supervised PCA

Incorporates class covariance structure into the PCA objective to bias principal components toward discriminative directions while retaining unlabeled variance information.

■ Semi-Supervised LDA (Linear Discriminant Analysis)

Uses unlabeled data to improve the within-class scatter estimate used in LDA, particularly valuable when the labeled set is small relative to the feature dimensionality.

■ Semi-Supervised Autoencoders

Adds a classification head to the autoencoder bottleneck; the supervised loss is applied only on labeled data while the reconstruction loss runs on all data, jointly learning a compact and class-relevant code.

■ Semi-Supervised t-SNE / UMAP

Constrains the t-SNE or UMAP embedding so that labeled points of the same class are embedded nearby, using the labels as soft repulsion / attraction constraints in the optimisation.

■ Semi-Supervised Metric Learning (SSML)

Learns a distance metric using labeled triplet / pair constraints (Siamese, triplet networks) while using unlabeled data to enforce global structure via graph or contrastive regularisation.

■ Deep Semi-Supervised Subspace Clustering (DSSSC)

Jointly learns a deep autoencoder and a self-expressive subspace clustering model, using labels to anchor certain subspace coefficients while clustering unlabeled data.

15. Federated & Privacy-Preserving Semi-Supervised Learning

Emerging methods that combine SSL with federated or privacy constraints, where labeled and unlabeled data reside on distributed clients.

■ FedMatch — Federated Semi-Supervised Learning

Distributes FixMatch-style SSL across federated clients. Introduces inter-client consistency loss and parameter decomposition (server shares partial parameters) to handle heterogeneous label distributions.

■ SemiFL — Semi-Supervised Federated Learning with Unlabeled Clients

Handles the extreme case where some clients have no labels at all. The server holds labeled data; unlabeled clients contribute via local SSL training and pseudo-label generation.

■ FedSemi — Communication-Efficient Federated SSL

Reduces communication cost in federated SSL by transmitting only sparse gradient updates or quantised model deltas, while maintaining SSL consistency across client rounds.

Quick-Reference Comparison Table

Family	Key Idea	Scalability	Label Efficiency	Best For
Generative Models	Model $P(X,Y)$	Medium	★★★	Structured data, NLP
Graph-Based	Label propagation on graph	Low–Med	★★★★	Network/relational data
Self-Training	Pseudo-labeling loop	High	★★	Any modality
Co-Training	Multi-view agreement	High	★★★	Multi-view data
Consistency Reg.	Prediction stability	High	★★★★	Vision, NLP, medical
Entropy Min.	Confident predictions	High	★★★	Classification
S3VM / TSVM	Max-margin + low-density	Low	★★★	Low-dim, tabular
SSL Clustering	Constrained clustering	Medium	★★	Exploration tasks
Active Learning	Optimal query selection	Medium	★★★★★	High annotation cost
Contrastive SSL	Representation learning	High	★★★★★	Vision, large-scale
SSL Regression	Smooth function estimation	Medium	★★★	Continuous outputs
Structured Pred.	Sequence/pixel labels	Medium	★★★	NLP, segmentation
Deep SSL (Modern)	Unified frameworks	High	★★★★★	SOTA benchmarks
Dim. Reduction	Label-guided embedding	Medium	★★★	Visualization, features
Federated SSL	Distributed + privacy	High	★★★	Privacy-critical apps

Key Takeaways & Further Reading

Semi-supervised learning has matured from simple self-training loops and graph-based propagation into a rich ecosystem of deep learning methods. Modern approaches such as FixMatch, MixMatch, and contrastive pretraining (SimCLR, DINO, MAE) achieve performance close to fully supervised baselines using only 1–10% of labels. For practitioners: **consistency regularisation methods are generally the most practical starting point** for computer vision; **pretrained language models (BERT, T5)** serve the equivalent role for NLP. Graph-based methods remain the tool of choice when relational structure is available. Active learning amplifies SSL gains by ensuring the labeled budget is spent on maximally informative examples.

Landmark Papers: Zhu (2005) survey; Chapelle, Schölkopf & Zien (2006) textbook; Oliver et al. (2018) NeurIPS realistic evaluation; Sohn et al. (2020) FixMatch; Berthelot et al. (2019) MixMatch; Chen et al. (2020) SimCLR; He et al. (2022) MAE; Assran et al. (2022) MSN.